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Course Name: Database Management Systems

Course Outcome

CO1-Understand the concept of database management systems and Relational database.
CO2- Identify the various data model used in database design.
CO3- Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4- Create and populate a RDBMS for a real-life application, with constraints and keys using SQL.
CO5- Select the information from a database by formulating complex queries in SQL.
CO6- Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7- Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8- Discuss recovery system and be familiar with introduction to web database, distributed databases.

Printed Pages: 3 **University Roll No. .....**

Mid Term Examination, Even Semester 2024-25
B.TECH. (CS/EC), Year: II, Semester: IV
BCSC 1003, Database Management Systems

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
1. Attempt all sections
2. Use only blue pen to answer.
3. No B sheet will be given.

Section – A

Attempt All Questions 3 X 5 = 15 Marks

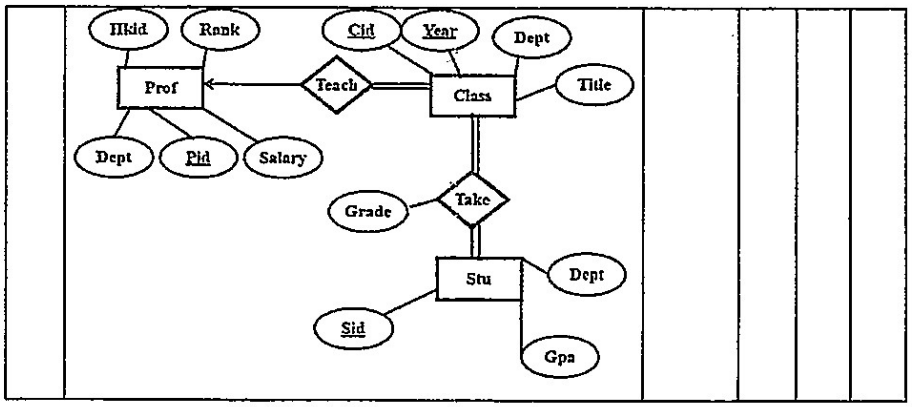
No.	Detail of Question	Marks	CO	BL	KL
1	A retail company stores all customer purchase records in text files. However, retrieving purchase history for a specific customer takes a long time due to the large volume of records. Would a file system or a DBMS be more efficient for this task, and why?	3	1	U	C
2	Write short note on: a. Primary key, Candidate key and Foreign key with the help of example. b. Domain constraint with the help of example.	3	3	E	F
3	A healthcare company is developing a patient record system. The project team holds meetings with **doctors, nurses, and administrative staff** to understand their data storage needs, such as patient history, medications, and billing. Which activity in the database development life cycle is the team currently performing? Give the other steps also that should be followed up for development of Database.	3	2	C	P
4	Describe data independence and its types with the help of diagram also give their significance in database design.	3	4	R	F

5	Consider the following schema: SUPPLIER (Sid, S_name, S_addr) PARTS (Pid, P_name, color) CATALOG (sid, pid, cost) Now, answer the following queries in Relational Algebra: (a) Find the name of all the suppliers who supply yellow parts. (b) Find the Sid of suppliers who supply every parts. (c) Find the Sid of suppliers who supply every red or green part.	3	3	E	P

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Suppose you are given relation R (A, B, C, D, E, F, G, H) and some functional dependency such as CH→G, A→BC, B→CHF, E→A, F→E Determine possible candidates keys up to two attributes set with explanation proof for the same.	5	4	E	C
7	Consider that a relation Student_Performance (Name, CourseNo, RollNo, Grade) has the following functional dependencies: {Name, CourseNo → Grade, RollNo, CourseNo → Grade, Name → RollNo, RollNo → Name} Determine the highest normal form of this relation.	5	6	E	C
8	Convert this ER Diagram into relations. How many relations we will get after conversion?	5	3	An	C


Hkid
Rank
Cid
Year
Dept
Prof
Teach
Class
Title
Dept
Pid
Salary
Grade
Take
Stu
Dept
Sid
Gpa